Smart Smishing Shield: An End-to-End SMS Spam Detection Framework using LSTM, DistilBERT and Adversarial Threat Analysis.

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Project Title: Smart Smishing Shield: An End-to-End SMS Spam Detection Framework using LSTM, DistilBERT transfer learning and Adversarial Threat Analysis.

Project Category: Deep Learning / Transfer Learning / Basic GEN AI

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1. Problem Definition

1.1 Business / Real-World Problem Statement

The rapid growth of mobile communication has led to an alarming increase in SMS spam, phishing, and smishing attacks. Fraudulent messages deceive mobile users, leading to financial loss, privacy breaches, and security vulnerabilities. Conventional filtering systems often fail to accurately classify such malicious messages, creating an urgent need for an advanced, reliable, and intelligent automated detection system to safeguard users from fraudulent mobile communication.

1.2 Project Objectives

- Understand the real-world problem.
- Analyze and preprocess the dataset.
- Build an appropriate AI model.
- Compare multiple experiments.
- Deploy the final solution.

1.3 Target Users / Stakeholders

Target Users:

- Mobile Phone Users / General Public: Individuals who are vulnerable to fraudulent SMS and smishing (SMS phishing) attacks and need protection against financial fraud.

- Telecom Operators & Service Providers: Organizations responsible for filtering network traffic and protecting their subscriber base from malicious messaging threats.
- Financial Institutions & Cybersecurity Teams: Entities looking to safeguard digital banking channels and mitigate fraud initiated through deceptive SMS vectors.

1.4 Input and Expected Output

Input:

- Raw SMS text messages (which may include normal text, adversarial spacing, or obfuscated characters).

Expected Output:

- Classification Result (Spam / Smishing vs. Safe Message)
- Confidence Score (Percentage value)
- Threat Level (High Risk / Low Risk)
- Actionable Security Recommendations (e.g., warning the user not to click links or share sensitive information if flagged as high risk)

1.5 Proposed AI Solution

The proposed AI solution leverages advanced Natural Language Processing (NLP) and Deep Learning via a fine-tuned DistilBERT transformer model. Unlike traditional rule-based filters that easily fail against modified text, this AI solution analyzes the deep contextual semantics of SMS messages. Combined with Whitespace Normalization and domain-specific engineered features, the model automatically detects subtle character obfuscation, malicious intent, and deceptive patterns in real-time, accurately classifying messages as safe or spam/smishing before they can harm the user.

1.6 Project Category

- Deep Learning
- NLP
- Transfer Learning
- Basic GEN AI

Technique / Model:

LSTM (Baseline), DistilBERT Transformer (Fine-tuned Model), & Gemini-generated Synthetic Data Augmentation

2. Dataset Selection & Understanding

2.1 Dataset Source

Dataset Source:

https://huggingface.co/datasets/notd5a/malicious-benign-sms-mms-dataset/blob/main/dataset_v3_for_deberta.csv

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2.2 Dataset Description

- Number of records / samples: Thousands of SMS/MMS text messages (268340)
- Number of features: Text content (SMS body) and target classification labels along with 26 features
- Number of classes: 2 Classes (0 for Safe, 1 for Spam)
- Dataset purpose: To train and evaluate deep learning models for accurate SMS threat detection
- Data source: Hugging Face (notd5a/malicious-benign-sms-mms-dataset)

2.3 Dataset Overview

Shape : (268340,26)Thousands of samples with core columns and engineered features

Info: Contains text data for messages and integer/float types for labels and numerical features. Memory usage is optimized, and missing values are handled during preprocessing.

Describe: Provides statistical insights on text length distributions and numerical feature ranges.

Observations:

- The dataset is well-structured for binary text classification tasks.
- Text samples include both standard messages and obfuscated texts requiring normalization.

2.4 Feature / Data Description

Feature	Data Type	Description
Message	Text	The raw SMS or MMS text content received by the user.
Label	Numeric	The target classification indicator (0 or 1)
engineered_features	Numeric	26 domain-specific extracted features capturing patterns, URLs, and character traits.

2.5 Target Variable / Class

Name: Label (or target)

Description: Binary classification indicator where 0 represents safe messages and 1 represents spam, phishing, or malicious smishing messages.

Prediction Goal: To accurately classify incoming SMS or MMS messages as either safe or spam/smishing to protect users from potential fraud.

3. Dataset & Preprocessing

- *Feature Separation:* The input data was structured cleanly by separating text content, numerical metadata features, and target output labels.
- *Data Splitting:* The dataset was split into training and testing sets, with 20% strictly reserved for validation.
- *Synthetic Data Augmentation:* To tackle data imbalance and scarcity of rare smishing variants, *Google Gemini* was utilized to generate high-quality synthetic text samples, significantly improving the model's training diversity and generalization.

4. Models Explored & Comparison

Multiple model architectures were built, trained, and rigorously compared:

1. *Model 1:*

Hybrid LSTM +Dense Neural Network

Model Type: Hybrid LSTM + Dense neural network combining text sequences with selected numerical features. Dataset: Original preprocessed SMS/Smishing dataset (N = 53,668 test samples). Optimization: Compiled using the Adam optimizer with a learning rate of 0.001, binary cross-entropy loss, and Early Stopping (patience=3, monitored via val_loss).

2. *Model 2:*

Model Name: Fine-Tuned DistilBERT Model (Transfer Learning)

Transfer Learning Concept: Transfer learning is a machine learning technique where a model developed for a task is reused as the starting point for a second task. In Natural Language Processing (NLP), this allows us to leverage pre-trained language models that already possess a deep understanding of grammar, semantics, and context from massive text , fine-tuning them on our specific SMS/Smishing dataset with fewer resources and high accuracy.

DistilBERT Architecture: DistilBERT is a small, fast, cheap, and light Transformer model based on the BERT architecture. By using knowledge distillation during the pre-training phase, DistilBERT reduces the size of a BERT model by 40% while retaining 97% of its language capabilities and being 60% faster.

Model Advantages for Smishing Detection: Unlike traditional recurrent networks, the attention mechanism in DistilBERT captures complex contextual dependencies and subtle phishing patterns in text messages, making it highly effective for binary classification between safe and spam/smishing messages.

3. *Model 3:* Gemini-augmented DistilBERT model.

Model Name: GenAI-Augmented DistilBERT Transfer Learning Model Dataset: The updated GenAI-augmented dataset combining the original preprocessed SMS/Smishing data with ~900 targeted synthetic messages to improve boundary-case robustness.

Architecture & Optimization: Utilizes the same robust transformer backbone and numerical feature integration pipeline, fine-tuned with optimized hyperparameters to minimize false positives on transactional and operational alerts.

Performance Goal: Assess whether the synthetic data effectively reduces false positives, improves boundary-case classification, and enhances overall security metrics compared to the original baseline.

Comprehensive Comparative Analysis

- **Baseline Evolution:** While the *Hybrid LSTM* served as a reliable baseline, it suffered from higher false positives on structural alerts and missed a higher number of spam instances.
- **Transfer Learning Leap:** Transitioning to *DistilBERT* dramatically improved the model's semantic comprehension, cutting False Negatives from 404 down to 215 and boosting the spam detection F1-score.
- **GenAI Data Augmentation Impact:** The final *GenAI-Augmented DistilBERT* model (combined with whitespace normalization) successfully targeted boundary-case misclassifications. It brought down False Positives significantly (from 372 in the baseline down to 249), ensuring that legitimate transactional alerts, login OTPs, and schedule reminders are safely processed without compromising high-risk smishing detection.

5. Technologies Used

- *Python* (Core programming language)
- *TensorFlow / Keras & KerasHub* (Deep learning framework)
- *DistilBERT & LSTM* (Neural network architectures)
- *Google Gemini* (Synthetic data augmentation)
- *Gradio* (Interactive web UI for deployment)
- *Google Colab* (Development and cloud execution environment)

6. Adversarial Threat Analysis & Edge Cases

- *Adversarial Threat Analysis:* Tested the model's resilience against adversarial text manipulation (e.g., character substitution, hidden Unicode characters, spaced-out scam words) to ensure attackers cannot easily bypass the filter.

7.Conclusion & Final Summary

- **Enhanced Threat Detection (Security Boost):**
  The **GenAI-Augmented DistilBERT** model demonstrates superior capability in capturing malicious spam and smishing messages compared to the baseline Hybrid LSTM and original DistilBERT models. Specifically, it achieves a massive reduction in **False Negatives** (dropping from 404 down to 188), ensuring that critical threats are rarely missed.
- **Superior Real-World Edge Case Handling:** When evaluated on critical everyday communications—such as meeting reminders, time-sensitive OTP codes, and shipping updates—the augmented model successfully prevents false alarms, classifying legitimate messages with up to 100% confidence whereas baseline models incorrectly flagged them as spam.
- **Acceptable Trade-Off & Future Scope:** While the aggressive spam sensitivity causes a slight increase in *False Positives* (safe messages incorrectly flagged), this is a strategically acceptable trade-off in security applications where missing actual spam is far more critical. Future improvements will focus on threshold tuning to minimize these minor false alarms while retaining high threat-detection accuracy.

8. Future Improvements

Handling Ambiguous Messages with More Data:

To further improve model robustness, we plan to train the model with more edge-case and ambiguous datasets—specifically normal messages that contain spam-like keywords—to reduce false positives and improve classification accuracy on tricky texts.

9. Installation

To set up the project locally or in a notebook environment, install the required dependencies:

pip install sci-kit learn matplotlib seaborn tensorflow keras keras-hub gradio